# HydroSOS Data Sharing Guidelines

## 1. Introduction to HydroSOS

WMO Member Countries consistently identify water-related challenges as one of their main priorities in the context of sustainable development, adaptation to climate change, and disaster risk reduction efforts. Hydrological information, including relevant status and outlook products, are required to address these challenges. In response to this need, the World Meteorological Organisation Hydrological Status (WMO) and Outlook System (HydroSOS) provides the first global operational mechanism for integrating hydrological status assessments and outlooks from, and for National Hydrological Services (NHS), in collaboration with River Basin Organisations and global modelling centres. HydroSOS is a global initiative led by the WMO and its partners to provide real-time information and forecasts on hydrological conditions, including droughts, floods, and soil moisture states. By offering data-driven insights, HydroSOS supports national and regional decision-makers in water resource management and disaster risk reduction.

The key objectives of HydroSOS include:

- Monitoring the current hydrological status at multiple spatial scales (national, regional, and global).
- Providing reliable short- to medium-term outlooks for hydrological conditions.
- Enhancing the capacity of National Meteorological and Hydrological Services (NMHSs) to develop and maintain operational hydrological monitoring systems.

HydroSOS integrates data from various sources, including river discharge, soil moisture, groundwater, and remotely sensed data, ensuring comprehensive monitoring and forecasting capabilities.

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## 2. Data Requirements for HydroSOS

HydroSOS relies on high-quality, timely datasets to support decision-making in water resource management. The following key data categories are required:

### 2.1 Streamflow (River Discharge)

- **Type:** Observational or modelled streamflow data.
- **Temporal Resolution:** Daily, monthly, seasonal.
- **Format:** CSV, GeoTIFF, JSON.
- **Metadata:** Catchment area, observation methods, and data quality.
- **Historical Data Requirement:** Minimum 20 years, preferably 30+ years (1991–2020 WMO standard).

### 2.2 Runoff

- **Type:** Observational or modeled runoff data.
- **Temporal Resolution:** Daily, monthly (primary), seasonal.
- **Spatial Resolution:** Point-based, catchment-based, or gridded.
- **Formats:** CSV, GeoTIFF, JSON.
- **Quality Control:** Includes completeness checks, gap filling, and uncertainty estimation.

### 2.3 Soil Moisture

- **Type:** Observational or modelled soil moisture data.
- **Temporal Resolution:** Daily, monthly (primary), seasonal.
- **Spatial Resolution:** Point-based, catchment-based, or gridded.
- **Formats:** CSV, JSON.
- **Quality Control:** Includes completeness checks and uncertainty estimation.

### 2.4 Groundwater

- **Type:** Observational or modelled groundwater data.
- **Temporal Resolution:** Monthly (mean) or seasonal (3-month cumulative).
- **Formats:** CSV, GeoTIFF, JSON.
- **Metadata:** Aligns with WMO best practices.
- **Data Quality:** Requires gap filling and completeness checks.

### 2.5 Cryosphere (Snow Water Equivalent - SWE)

- **Type:** Observational, model-based, or satellite-derived.
- **Temporal Resolution:** Weekly, bi-weekly, monthly.
- **Formats:** CSV
- **Quality Control:** 80% completeness, flagged gaps, uncertainty estimation.
- **Geographic Coverage:** Global (wherever snow exists), regional, or national.

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**3. Data Sharing Mechanisms**

To facilitate seamless data exchange, HydroSOS supports the following delivery methods:

- **API:** Automated data sharing.
- **FTP:** File transfer protocols for bulk uploads.
- **SharePoint:** Secure cloud storage and sharing.
- **WHOS:** Future integration with the WMO Hydrological Observing System (WHOS).

**Standardised Formats & Protocols:**

- Data should comply with WMO, Global Runoff Data Centre (GRDC), and WIGOS standards.
- Metadata must be provided with each dataset to ensure usability and interoperability.

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**4. Data Quality and Reliability**

Ensuring data reliability is critical for the success of HydroSOS. Recommended practices include:

- **Completeness:** Records should cover **at least 80% of the period of interest.**
- **Quality Control:** Initial QC checks, gap filling, and outlier flagging should be conducted.
- **Metadata:** Must include data sources, collection methods, and quality control procedures.
- **Uncertainty Estimation:** Data providers should describe associated uncertainties (e.g., standard deviations, confidence intervals).

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**5. Guidelines for Data Contributors**

**Step 1: Identify Relevant Data**

- Ensure your data meets HydroSOS specifications (streamflow, soil moisture, groundwater, cryosphere data).

**Step 2: Format Data Correctly**

- Use standardised formats like CSV, and include metadata adhering to WMO standards.

**Step 3: Ensure Data Quality**

- Conduct quality control, including gap filling and uncertainty estimation.

**Step 4: Share Data Securely**

- Use approved delivery methods: API, FTP, SharePoint.

**Step 5: Maintain Data Updates**

- Provide regular updates (e.g., monthly or seasonal) with minimal latency (within 7 days of the observation period).

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**6. Expected Outcomes of HydroSOS Participation**

By contributing to and utilising HydroSOS, NMHSs and stakeholders can:

- Improve decision-making in water resource management.
- Enhance early warning systems for droughts and floods.
- Strengthen collaborations between national and international hydrological communities.
- Gain access to a global hydrological monitoring system for better forecasting.

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**7. Conclusion**

HydroSOS represents a transformative approach to hydrological monitoring and forecasting. By adhering to the data sharing and quality guidelines outlined in this document, NMHSs and partners can play a crucial role in advancing global water security and resilience.

For further information or technical support, contact the HydroSOS coordination team or refer to WMO guidelines.